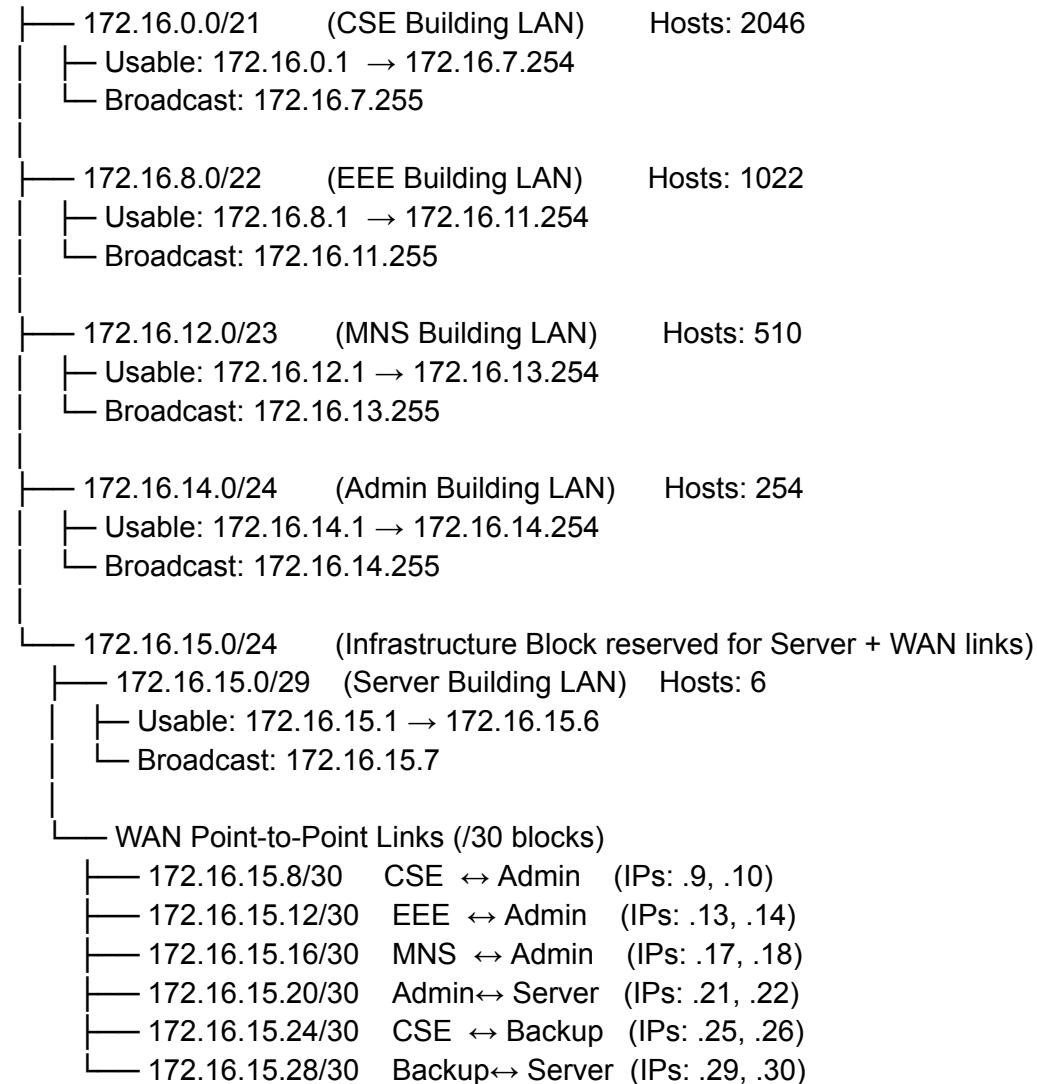


VLSM TREE (Base Block: 172.16.0.0/16)

172.16.0.0/16



IP ADDRESSING TABLE (VLSM/Lan)

Building	Users	Subnet Size	CIDR	Subnet Mask	Network Address	Gateway IP	Last Usable IP	Broadcast
CSE	1500	2048 IPs	/21	255.255.248.0	172.16.0.0	172.16.0.1	172.16.7.254	172.16.7.255
EEE	700	1024 IPs	/22	255.255.252.0	172.16.8.0	172.16.8.1	172.16.11.254	172.16.11.255
MNS	500	512 IPs	/23	255.255.254.0	172.16.12.0	172.16.12.1	172.16.13.254	172.16.13.255
Admin	250	256 IPs	/24	255.255.255.0	172.16.14.0	172.16.14.1	172.16.14.254	172.16.14.255
Server	5	8 IPs	/29	255.255.255.248	172.16.15.0	172.16.15.1	172.16.15.6	172.16.15.7

WAN LINK ADDRESSING TABLE (/30)

Link	Network Address	CIDR	Subnet Mask	Router 1 IP	Router 2 IP
CSE ↔ Admin	172.16.15.8	/30	255.255.255.252	172.16.15.9	172.16.15.10
EEE ↔ Admin	172.16.15.12	/30	255.255.255.252	172.16.15.13	172.16.15.14
MNS ↔ Admin	172.16.15.16	/30	255.255.255.252	172.16.15.17	172.16.15.18
Admin ↔ Server	172.16.15.20	/30	255.255.255.252	172.16.15.21	172.16.15.22
CSE ↔ Backup	172.16.15.24	/30	255.255.255.252	172.16.15.25	172.16.15.26
Backup ↔ Server	172.16.15.28	/30	255.255.255.252	172.16.15.29	172.16.15.30

- Lines starting with # are comments for documentation.
EEE ROUTER CONFIGURATION

```
#####  
enable  
conf t  
hostname EEE_Router  
  
# EEE LAN Interface  
interface g0/0  
ip address 172.16.8.1 255.255.252.0  
no shutdown  
exit  
  
# EEE ↔ Admin WAN (EEE is DCE)  
interface s0/0/0  
ip address 172.16.15.13 255.255.255.252  
clock rate 64000  
no shutdown  
exit  
  
# DHCP Relay for EEE LAN  
interface g0/0  
ip helper-address 172.16.15.5  
exit  
  
end  
wr
```

```
#####  
# MNS ROUTER CONFIGURATION  
#####  
enable  
conf t  
hostname MNS_Router  
  
# MNS LAN Interface  
interface g0/0  
ip address 172.16.12.1 255.255.254.0  
no shutdown  
exit  
  
# MNS ↔ Admin WAN (MNS is DCE)  
interface s0/0/0  
ip address 172.16.15.17 255.255.255.252  
clock rate 64000
```

```
no shutdown
exit
```

```
# DHCP Relay for MNS LAN
interface g0/0
ip helper-address 172.16.15.5
exit
```

```
end
wr
```

```
#####
# CSE ROUTER CONFIGURATION
# (Recovered and Fixed)
#####
enable
conf t
hostname CSE_Router
```

```
# CSE LAN Interface
interface g0/0
ip address 172.16.0.1 255.255.248.0
no shutdown
exit
```

```
# CSE ↔ Admin WAN (CSE is DCE)
interface s0/0/0
ip address 172.16.15.9 255.255.255.252
clock rate 64000
no shutdown
exit
```

```
end
wr
```

```
#####
# ADMIN ROUTER CONFIGURATION
#####
enable
conf t
hostname Admin_Router
```

```
# Admin LAN Interface
interface g0/0
ip address 172.16.14.1 255.255.255.0
no shutdown
exit

# Admin ↔ CSE (Admin is DTE)
interface s0/0/0
ip address 172.16.15.10 255.255.255.252
no shutdown
exit

# Admin ↔ EEE (Admin is DTE)
interface s0/0/1
ip address 172.16.15.14 255.255.255.252
no shutdown
exit

# Admin ↔ MNS (Admin is DTE)
interface s0/1/0
ip address 172.16.15.18 255.255.255.252
no shutdown
exit

# Admin ↔ Server (Admin is DTE)
interface s0/1/1
ip address 172.16.15.21 255.255.255.252
no shutdown
exit

end
wr
```

```
#####
# SERVER ROUTER CONFIGURATION
#####
enable
conf t
hostname Server_Router

# Server LAN Interface
interface g0/0
ip address 172.16.15.1 255.255.255.248
```

```
no shutdown
exit
```

```
# Server ↔ Admin WAN (Server is DTE)
interface s0/0/0
ip address 172.16.15.22 255.255.255.252
no shutdown
exit
```

```
end
wr
```

```
# DHCP Server CSE_Pool configuration
```

Pool Name: CSE_Pool

Default Gateway: 172.16.0.1

DNS Server: 172.16.15.5

Start IP Address: 172.16.0.2

Subnet Mask: 255.255.248.0 (/21)

Maximum no. of users = 1500

Excluded IPs: 172.16.7.254

```
# DHCP Server EEE_Pool configuration
```

Pool Name: EEE_Pool

Default Gateway: 172.16.8.1

DNS Server: 172.16.15.5

Start IP Address: 172.16.8.2

Subnet Mask: 255.255.252.0 (/22)

Maximum no. of users = 800

Excluded IPs: 172.16.11.254

Pool Name: MNS_Pool

Default Gateway: 172.16.12.1

DNS Server: 172.16.15.5

Start IP Address: 172.16.12.2

Subnet Mask: 255.255.254.0 (/23)

Maximum no. of users = 100

Excluded IPs: 172.16.13.254

#Supercomputer IP configurations

Building	Super Computer Static IP	Subnet Mask	Default Gateway
CSE	172.16.7.254	255.255.248.0	172.16.0.1
EEE	172.16.11.254	255.255.252.0	172.16.8.1
MNS	172.16.13.254	255.255.254.0	172.16.12.1
Admin	172.16.14.254	255.255.255.0	172.16.14.1
Server	172.16.15.6	255.255.255.248	172.16.15.1

DNS server for all supercomputers - 172.16.15.5

Rip configuration for all 3 servers (same commands)

```
enable
configure terminal
router rip
version 2
no auto-summary
network 172.16.0.0
exit
```

Static Routing for Server Building

```
enable
configure terminal
ip route 0.0.0.0 0.0.0.0 172.16.15.21
```

exit

Admin Router Configuration

```
enable
configure terminal
ip route 172.16.15.0 255.255.255.248 172.16.15.22
router rip
redistribute static
exit
```

Helper ip configuration for all routers (same commands)

```
enable
configure terminal
interface g0/0
ip helper-address 172.16.15.5
exit
```

###WEB SERVER & DNS CONFIG

new content of index.html = <h1>Welcome To Brac University</h1>

toggled DNS service to ON

name = www.bracu.com

type = A record

address = 172.16.15.5

###EMAIL SERVER SETUP

Domain name = mail.com

User: cse, Password: 123

User: eee, Password: 123

User: mns, Password: 123

User: admin, Password: 123

###SUPERCOMPUTER EMAIL CONFIG

for cse -

cse
cse@mail.com
172.16.15.5
172.16.15.5
cse
123

for eee -

eee
eee@mail.com
172.16.15.5
172.16.15.5
eee
123

for mns -

mns
mns@mail.com
172.16.15.5
172.16.15.5
mns
123

for admin -

admin
admin@mail.com
172.16.15.5
172.16.15.5
admin
123

###SECURITY

enable
configure terminal

```
ip access-list extended ISOLATION_ACL
permit ip any 172.16.15.0 0.0.0.7
```

```
deny ip 172.16.0.0 0.0.7.255 172.16.8.0 0.0.3.255
```

```
deny ip 172.16.0.0 0.0.7.255 172.16.12.0 0.0.1.255
```

```
deny ip 172.16.8.0 0.0.3.255 172.16.12.0 0.0.1.255
```

```
permit ip any any
```

```
exit
```

```
interface s0/0/0
```

```
interface s0/0/1
```

```
ip access-group ISOLATION_ACL in
```

```
interface s0/1/0
```

```
ip access-group ISOLATION_ACL in
```

###BACKUP ROUTER CONFIG

```
enable
```

```
configure terminal
```

```
hostname Backup_Router
```

```
interface s0/0/0
```

```
ip address 172.16.15.26 255.255.255.252
```

```
no shutdown
```

```
exit
```

```
interface s0/0/1
```

```
ip address 172.16.15.29 255.255.255.252
```

```
no shutdown
```

```
exit
```

```
ip route 172.16.0.0 255.255.248.0 172.16.15.25
```

```
ip route 172.16.15.0 255.255.255.248 172.16.15.30
```

exit

enable
configure terminal
interface s0/0/0
ip address 172.16.15.26 255.255.255.252
no shutdown
exit

###CSE ROUTER UPDATE

enable
configure terminal

ip route 172.16.15.0 255.255.255.248 172.16.15.26 130

exit

enable
configure terminal
interface s0/0/1
ip address 172.16.15.25 255.255.255.252
no shutdown
exit

###SERVER ROUTER UPDATE

enable
configure terminal

ip route 172.16.0.0 255.255.248.0 172.16.15.29 130
exit

enable
configure terminal
interface s0/0/1
ip address 172.16.15.30 255.255.255.252
no shutdown
exit

no cli for secret mail
#Isolation of cse,eee,mns

```
enable
conf t
```

```
ip access-list extended ISOLATION_ACL
permit ip any 172.16.15.0 0.0.0.7
deny ip 172.16.0.0 0.0.7.255 172.16.8.0 0.0.3.255
deny ip 172.16.0.0 0.0.7.255 172.16.12.0 0.0.1.255
deny ip 172.16.8.0 0.0.3.255 172.16.12.0 0.0.1.255
permit ip any any
exit
```

```
interface s0/0/0
ip access-group ISOLATION_ACL in
exit
```

```
interface s0/0/1
ip access-group ISOLATION_ACL in
exit
```

```
interface s0/1/0
ip access-group ISOLATION_ACL in
exit
```

```
end
wr
```